

One of the crucial factors in this phenomenon is the distance between the two viewpoints. Too great a distance would make the dual images too dissimilar to be overlaid by our brain. Along with the all important dual image, there are a number of features that can create a sense of depth even with a single image, like perspective lines, motion parallax (distant objects appearing to move slower), surface textures varying over distance. They are the reason why, when you close one of your eyes, you aren't suddenly greeted with a flat-depthless world.

Types of 3D

Even though the basic principles might be the same, the way 3D has been implemented over the years has been drastically varied. Some of these methods heavily relied on figuring out the role hardware has to play in the experience. Even something like a Whereas some of the other methods used software as the powering tool. Let us look at both in more details.

The Tools:

- 1) Stereo displays** - All about implementing those two important images in front of the left and the right eye, even this tool consists of number of variations
 - a) **Side by Side image** - The kind that we were explaining earlier. Two slightly different images side by side creating one image. This method is prone to being uncomfortable without visual aids. A significant present day example of this is a Head Mounted Display - a helmet like apparatus with stereo displays which can show pictures, films and games. Due to being placed much closer to the eye, a wider field of view is possible. We'll cover more on this in the next chapter
 - b) **Anaglyph Display** - This type of imaging superimposes two coloured images which are differently filtered. For viewing, each eye uses a chromatically different, and usually opposite, filter and hence is able to only see the image meant for that eye. For ease of production, red and blue are usually the colour of choice, although cyan can achieve a more accurate colour reproduction.
 - c) **Polarization Systems** - Without delving into highly conceptual physics concept, let us briefly tell you about polarization. Any physical wave which can oscillate in more than one orientation (read 45° , 180° etc) has the property of polarization which defines that orientation. This property of light waves is used to superimpose two images of different polarizations